

DEPARTMENT OF MATHEMATICS AND APPLIED MATHEMATICS

Mathematics III

Topics in Analysis (3TN)

Linear Continuous Mappings

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0.1 Introduction

Definition 0.1

- (i) **(Linear mapping)** A mapping $L : V \rightarrow W$ between two vector spaces V and W over the same field $\mathbb{K}$ is said to be a *linear operator* or a *linear mapping* if $L(\alpha u + \beta v) = \alpha L(u) + \beta L(v)$ for every $u, v \in V$ and every $\alpha, \beta \in \mathbb{K}$. We will use the notation sometimes Lu in place of $L(u)$.
- (ii) **(Isomorphism)** $L : V \rightarrow W$ is called isomorphism if L is linear and bijective.
- (iii) **(Kernel)** We call *kernel* of the linear operator $L : V \rightarrow W$ the subspace of V defined by

$$\text{Ker}(L) := \{v \in V : L(v) = 0_W\}.$$

- (iv) **(Range)** We call *range* of the linear operator $L : V \rightarrow W$ the subspace of W defined by

$$\mathcal{R}(L) := \{w \in W : \exists v \in V \text{ such that } w = L(v)\}.$$

Remark 0.2 (Motivation for the study of linear continuous mappings)

- (i) In general, linear operators are defined in a proper subspace of the given linear space V . In most of the applications, the domain $D(L)$ of the operator L is a dense subspace of V . However, we will confine ourselves in these lectures to operators defined in all the vector space V .
- (ii) As we will see later, linear mappings between normed spaces are not always continuous except on finite dimensional spaces. In fact, if $\dim(V) = n$ and $\dim(W) = m$ with $(e_i)_{i=1}^n$ and $(f^j)_{j=1}^m$ being bases of V and W respectively, then we will have for $L : V \rightarrow W$ linear that

$$\begin{aligned} \forall x \in V \quad x &= \sum_{i=1}^n x_i e_i, \quad \forall i \quad L(e_i) = \sum_{j=1}^m a_i^j f^j \\ \Rightarrow \forall x \in V \quad &\begin{cases} L(x) = L\left(\sum_{i=1}^n x_i e_i\right) = \sum_{i=1}^n x_i L(e_i) \\ = \sum_{i=1}^n \sum_{j=1}^m a_i^j x_i f^j = \sum_{j=1}^m \sum_{i=1}^n a_i^j x_i f^j \end{cases} \end{aligned}$$

We then obtain a linear continuous mapping $\Phi : \mathbb{R}^n \rightarrow \mathbb{R}^m$ defined by

$$\Phi(x_1, x_2, \dots, x_n) = \left(\sum_{i=1}^n a_i^1 x_i, \sum_{i=1}^n a_i^2 x_i, \dots, \sum_{i=1}^n a_i^m x_i \right).$$

- (ii) On infinite dimensional spaces there are linear mappings which are continuous. For instance, we can take $V := \mathcal{C}^1([0, 1], \mathbb{R})$ and $W := \mathcal{C}([0, 1], \mathbb{R})$ both equipped with norm $\|\cdot\|_\infty$. The mapping $T : V \rightarrow W$ defined by $T(f) = f'$, for every $f \in V$, is linear but not continuous. In fact, we can take $f_n(x) = \frac{1}{n+1}x^{n+1}$ for every $x \in [0, 1]$. It follows that $f'_n(x) = x^n$ and $\|f_n\|_\infty = \frac{1}{n+1} \rightarrow 0$ while $\|f'_n\|_\infty = 1$. Similarly, one could also take the example $f_n(x) = \frac{\sin(nx)}{n}$ for which $f'_n(x) = \cos(nx)$ and hence, $\|f_n\|_\infty = \frac{1}{n} \rightarrow 0$ while $\|f'_n\|_\infty = 1$.
- (iii) On the other hand, even when the linear mapping invertible mapping is continuous, its inverse mapping though being linear is not always continuous. In fact, we can construct an example of a linear continuous invertible mapping whose inverse is the mapping in the example in (ii). We consider the mapping $L : W \rightarrow V$ defined by

$$L(f)(x) := \int_0^x f(t)dt \quad \forall x \in [0, 1], \quad \forall f \in W.$$

The mapping L is linear and

$$|L(f)(x)| \leq \int_0^x |f(t)|dt \leq x \|f\|_\infty \leq \|f\|_\infty \quad \Rightarrow \quad \|L(f)\|_\infty = \sup_{x \in [0, 1]} |L(f)(x)| \leq \|f\|_\infty.$$

The mapping L is also injective. In fact, if $f \in \text{Ker}(L)$, we get $L(f)(x) = 0$ for every $x \in [0, 1]$ and hence,

$$\int_0^x f(t)dt = 0 \quad \forall x \in [0, 1] \quad \xRightarrow[\text{Differentiating}]{} \quad f(x) = 0 \quad \forall x \in [0, 1] \quad (\text{By (FTC)})$$

Therefore, $L : W \rightarrow \text{Im}(L) = \tilde{V} := \{f \in V : f(0) = 0\}$ is invertible with inverse being the mapping T in the example in (ii), which is not continuous as shown in (ii).

Definition 0.3 (Linear continuous mapping) Let $(V, \|\cdot\|_V)$ and $(W, \|\cdot\|_W)$ be two normed spaces over the same field $\mathbb{K}$. The linear mapping $L : V \rightarrow W$ is continuous if L is a continuous function between the metric spaces (V, d_V) and (W, d_W) where d_V and d_W are the distances induced by the norms $\|\cdot\|_V$ and $\|\cdot\|_W$ respectively.

The following proposition gives some characterizations of linear continuous mappings.

Proposition 0.4 Let $(V, \|\cdot\|_V)$ and $(W, \|\cdot\|_W)$ be two normed spaces. Let $L : V \rightarrow W$ be a linear mapping. Then the following are equivalent:

- (i) the mapping L is continuous;
- (ii) the mapping L is continuous at 0_V ;
- (iii) there exists $C > 0$ such that $\|Lv\|_W \leq C\|v\|_V$ for any $v \in V$;
- (iv) the mapping L is a Lipschitz function.

Proof: (i) $\Rightarrow$ (ii) is obvious.

(ii) $\Rightarrow$ (iii). Notice that (iii) holds for $v = 0_V$. If L is continuous at 0_V , then for $\varepsilon = 1$, there exists $\delta > 0$ such that $\|u\|_V < \delta \Rightarrow \|Lu\|_W < 1$.

Let $v \in V \setminus \{0_V\}$. For $u = \frac{\delta}{2} \frac{v}{\|v\|_V}$, we get that $\|u\| = \frac{\delta}{2} < \delta$ and hence

$$1 > \|Lu\|_W = \left\| L\left(\frac{\delta}{2} \frac{v}{\|v\|_V}\right) \right\|_W = \left\| \frac{\delta}{2\|v\|_V} L(v) \right\|_W = \frac{\delta}{2\|v\|_V} \|L(v)\|_W.$$

So, we obtain that $\|Lv\|_W \leq C \|v\|_V$ with $C := \frac{2}{\delta}$.

(iii) $\Rightarrow$ (iv). Let $u_1, u_2 \in V$ then writing (iii) for $v = u_1 - u_2$ and using the linearity of L we find,

$$\|Lu_1 - Lu_2\|_W = \|L(u_1 - u_2)\|_W \leq C \|u_1 - u_2\|_V.$$

(iv) $\Rightarrow$ (i) is obvious. □

Proposition 0.5 Let $(V, \|\cdot\|_V)$ and $(W, \|\cdot\|_W)$ be two normed spaces. Let $\mathcal{L}(V, W)$ be the vector space of linear continuous mapping from V to W . Set

$$\|L\|_{\mathcal{L}(V, W)} := \sup_{v \neq 0_V} \frac{\|Lv\|_W}{\|v\|_V} \quad \forall L \in \mathcal{L}(V, W).$$

Then the following facts hold:

- (a) $\|\cdot\|_{\mathcal{L}(V, W)}$ is a norm on $\mathcal{L}(V, W)$.
- (b) $\|L\|_{\mathcal{L}(V, W)} = \sup_{\|v\|_V=1} \|Lv\|_W = \sup_{\|v\|_V \leq 1} \|Lv\|_W$.

Proof: (a) Let us show that $\|\cdot\|_{\mathcal{L}(V, W)}$ is a norm on $\mathcal{L}(V, W)$.

- (i) $\|L\|_{\mathcal{L}(V, W)} \geq 0 \quad \forall L \in \mathcal{L}(V, W)$ and

$$\|L\|_{\mathcal{L}(V, W)} = 0 \quad \Leftrightarrow \quad L(v) = 0_W \quad \forall v \in V \text{ with } v \neq 0_V.$$

Since, $L(0_V) = 0_W$ it follows that $L(v) = 0$ for every $v \in V$ and hence $L = 0_{\mathcal{L}(V, W)}$.

- (ii) Let $L \in \mathcal{L}(V, W)$ and let $\lambda \in \mathcal{K}$. We find that

$$\|\lambda L\|_{\mathcal{L}(V, W)} = \sup_{v \neq 0_V} \frac{\|(\lambda L)(v)\|_W}{\|v\|_V} = \sup_{v \neq 0_V} \frac{|\lambda| \|L(v)\|_W}{\|v\|_V} = |\lambda| \sup_{v \neq 0_V} \frac{\|L(v)\|_W}{\|v\|_V} = |\lambda| \|L\|_{\mathcal{L}(V, W)}.$$

- (iii) Let $L_1, L_2 \in \mathcal{L}(V, W)$. For every $v \in V$ with $v \neq 0_V$, we have that

$$\begin{aligned} \frac{\|(L_1 + L_2)v\|_W}{\|v\|_V} &= \frac{\|L_1(v) + L_2(v)\|_W}{\|v\|_V} \leq \frac{\|L_1(v)\|_W}{\|v\|_V} + \frac{\|L_2(v)\|_W}{\|v\|_V} \\ &\leq \sup_{v \neq 0_V} \frac{\|L_1(v)\|_W}{\|v\|_V} + \sup_{v \neq 0_V} \frac{\|L_2(v)\|_W}{\|v\|_V} = \|L_1\|_{\mathcal{L}(V, W)} + \|L_2\|_{\mathcal{L}(V, W)}. \end{aligned}$$

Now, taking the supremum we get that $\|L_1 + L_2\|_{\mathcal{L}(V, W)} \leq \|L_1\|_{\mathcal{L}(V, W)} + \|L_2\|_{\mathcal{L}(V, W)}$.

Now set

$$\alpha := \sup_{\|v\|_V=1} \|Lv\|_W \quad \text{and} \quad \beta := \sup_{\|v\|_V \leq 1} \|Lv\|_W.$$

We want to show that for every $L \in \mathcal{L}(V, W)$ we have that $\|L\|_{\mathcal{L}(V, W)} = \alpha = \beta$.

First of all, notice that

$$\alpha \leq \|L\|_{\mathcal{L}(V, W)} \quad \text{and} \quad \alpha \leq \beta.$$

We are going to show that

$$\|L\|_{\mathcal{L}(V, W)} \leq \alpha \quad \text{and} \quad \beta \leq \|L\|_{\mathcal{L}(V, W)}.$$

Let $v \in V$ and set $u := \frac{v}{\|v\|_V}$ for which it follows that $\|u\|_V = 1$. Hence,

$$\underbrace{\frac{\|L(v)\|_W}{\|v\|_V} = \left\| \frac{1}{\|v\|_V} L(v) \right\|_W}_{\text{property of } \|\cdot\|_W} = \underbrace{\left\| L\left(\frac{v}{\|v\|_V}\right) \right\|_W}_{\text{Linearity of } L} = \|L(u)\|_W \leq \alpha.$$

Taking the supremum, we get that $\|L\|_{\mathcal{L}(V, W)} \leq \alpha$.

On the other hand, Let $v \in V$ with $0 < \|v\|_V \leq 1$. We get that

$$\|L(v)\|_W = \frac{\|L(v)\|_W}{1} \leq \frac{\|L\|_W}{\|v\|_V} \leq \|L\|_{\mathcal{L}(V, W)}.$$

Now taking the supremum, we get that $\beta \leq \|L\|_{\mathcal{L}(V, W)}$ and this completes the proof the proposition. ■

Example 0.6 Let $V = \mathcal{C}([0, 1], \mathbb{R})$ be the vector over $\mathbb{R}$ of continuous functions from $[0, 1]$ to $\mathbb{R}$, equipped with the norm $\|f\|_\infty := \sup_{x \in [0, 1]} |f(x)|$. Consider the mapping $L : V \rightarrow \mathbb{R}$ defined by

$$L(f) := f(1/3) \quad \forall f \in V.$$

Show that $L \in \mathcal{L}(V, \mathbb{R})$ with $\|L\|_{\mathcal{L}(V, \mathbb{R})} = 1$.

Solution: Let us first show that the mapping L is linear. To this end, let $f, g \in V$ and let $\alpha, \beta \in \mathbb{R}$.

$$L(\alpha f + \beta g) = (\alpha f + \beta g)(1/3) = \alpha f(1/3) + \beta g(1/3) = \alpha L(f) + \beta L(g).$$

Now,

$$L \in \mathcal{L}(V, \mathbb{R}) \text{ with } \|L\|_{\mathcal{L}(V, \mathbb{R})} \leq 1 \quad (*) \quad \Leftrightarrow \quad |L(f)| \leq \|f\|_\infty \quad \forall f \in V \quad (**).$$

It is important for you to explain why the equivalence is true!!!

So, let us prove that $(**)$ in the yellow box holds. This is trivial from the definition of the norm $\|\cdot\|_\infty$. In fact, $|L(f)| = |f(1/3)| \leq \sup_{x \in [0, 1]} |f(x)| = \|f\|_\infty$.

Now, to complete the solution of this exercise, it remains to show that $\|L\|_{\mathcal{L}(V, \mathbb{R})} \geq 1 \quad (***)$.

Now, from the definition of

$$\|L\|_{\mathcal{L}(V, \mathbb{R})} = \sup_{\|f\|_\infty=1} |L(f)| \quad (\text{as supremum})$$

we will obtain $(***)$ if we find a function $f \in V$ such that $\|f\|_\infty = 1$ and $|L(f)| = 1$. **(Think about this!)**. Let us find such a function f . In this case, it is easy to find such function. Just take the function f identically equal to 1 and get that

$$\|f\|_\infty = 1 \quad \text{and} \quad |L(f)| = |f(1/3)| = 1.$$

Hence, combining $(*)$ and $(***)$ we get that $L \in \mathcal{L}(V, \mathbb{R})$ with $\|L\|_{\mathcal{L}(V, \mathbb{R})} = 1$.

Remark 0.7 Notice, if you were asked to show only that $L \in \mathcal{L}(V, \mathbb{R})$ then you would need to show that L is linear and then you would seek a number $C > 0$ such that

$$|L(v)| \leq C \|f\|_\infty \quad \forall f \in V.$$

Theorem 0.8 Let $(V, \|\cdot\|_V)$ and $(W, \|\cdot\|_W)$ be two normed spaces. Let $\mathcal{L}(V, W)$ be the vector space of linear continuous mapping from V to W equipped with the norm

$$\|L\|_{\mathcal{L}(V, W)} := \sup_{\|v\|_V \leq 1} \|Lv\|_W \quad \forall L \in \mathcal{L}(V, W).$$

If $(W, \|\cdot\|_W)$ is a Banach space, then also $(\mathcal{L}(V, W), \|\cdot\|)$ is a Banach space.

Proof: Let $\{L_n\}$ be a Cauchy sequence in $\mathcal{L}(V, W)$, i.e., for every $\epsilon > 0$ there exist $n_\epsilon \in \mathbb{N}$ such that $\|L_n - L_m\|_{\mathcal{L}(V, W)} < \epsilon$ for any $n, m \geq n_\epsilon$; hence from the definition of $\|\cdot\|_{\mathcal{L}(V, W)}$, we get for any $u \in V$,

$$\|L_n u - L_m u\|_W \leq \epsilon \|u\|_V \quad \forall n, m \geq n_\epsilon. \quad (1)$$

Therefore for any fixed $u \in V$, the sequence $\{L_n u\}$ is a Cauchy sequence in W and since W is a Banach space, it has a limit that we denote by Lu . We obtain then a mapping $L : V \rightarrow W$ that easily turns out to be linear. In fact, for any $\alpha, \beta \in \mathbb{K}$ and any $u, v \in V$, we have that

$$L_n(\alpha u + \beta v) = \alpha L_n u + \beta L_n v \quad \forall n \in \mathbb{N}$$

and passing to limit on both sides when $n \rightarrow \infty$ we get that $L(\alpha u + \beta v) = \alpha Lu + \beta Lv$. Moreover, for any $u \in V$ we have

$$\begin{aligned} \|Lu\|_W &= \lim_{n \rightarrow \infty} \|L_n u\|_W \leq \lim_{n \rightarrow \infty} [\|L_n u - L_{n_\epsilon} u\|_W + \|L_{n_\epsilon} u\|_W] \\ &\leq \liminf_{n \rightarrow \infty} [\|L_n - L_{n_\epsilon}\|_{\mathcal{L}(V, W)} \|u\|_W + \|L_{n_\epsilon} u\|_W] \quad (\text{Explain why } \liminf_{n \rightarrow \infty}) \\ &\leq [\epsilon + \|L_{n_\epsilon}\|_{\mathcal{L}(V, W)}] \|u\|_V \end{aligned}$$

for which it follows that L is a bounded operator. Passing to the limit in (1) we get

$$\|L_n u - Lu\|_W = \lim_{m \rightarrow \infty} \|L_m u - L_m u\|_W \leq \epsilon \|u\|_V \quad \forall n \geq n_\epsilon$$

from which it follows that $\{L_n\}$ converges to L in $\mathcal{L}(V, W)$. □

Remark 0.9 Any linear continuous mapping $L : (V, \|\cdot\|_V) \rightarrow (\mathbb{K}, |\cdot|)$ is called a *continuous linear form* and we use the notation $\mathcal{L}(V, \mathbb{K}) = V'$ which is called the *dual space* of V . If $L \in V'$ then we use the notation $\langle L, v \rangle = L(v) = Lv$ where $\langle \cdot, \cdot \rangle$ are called the duality brackets between V' and V .

From Theorem 0.8, it follows that V' is always a Banach space.

What is the definition of the norm on V' ?

Exercise 0.10 Let $1 \leq p < \infty$ and let q be the exponent conjugate to p , i.e., $\frac{1}{p} + \frac{1}{q} = 1$ with the convention that $q = \infty$ when $p = 1$. Let $y = (y_n) \in \ell^q(\mathbb{C})$ and let $L_y : \ell^p(\mathbb{C}) \rightarrow \mathbb{C}$ be defined by

$$L_y(x) := \sum_{n=0}^{\infty} x_n y_n \quad \forall x = (x_n) \in \ell^p(\mathbb{C}).$$

Then show that $L_y \in \mathcal{L}(\ell^p(\mathbb{C}), \mathbb{C})$ (i.e., L_y is linear and continuous) with $\|L_y\|_{\mathcal{L}(\ell^p(\mathbb{C}), \mathbb{C})} = \|y\|_q$.

Example 0.11

If $V = \mathbb{R}^n$, the dual $(\mathbb{R}^n)'$ is isomorphic and isometric to $\mathbb{R}^n$. Indeed, if $(e_1, e_2, \dots, e_n)$ is the canonical basis of $\mathbb{R}^n$, any linear form L is represented as

$$Lu = \sum_{i=1}^n a_i u_i \quad \text{with } a := (Le_1, Le_2, \dots, Le_n)$$

and hence

$$\|L\|_{(\mathbb{R}^n)'} = \|a\|_{\mathbb{R}^n}.$$

Exercise 0.12 Let V, W be Banach spaces. Verify that $V \times W$ equipped with the norm

$$\|(v, w)\|_{V \times W} := \|v\|_V + \|w\|_W,$$

is a Banach space. Prove that $(V \times W)'$ is isomorphic to $V' \times W'$.

Exercise 0.13 Let

$$c_0 := \left\{ x = \{x_n\} : \lim_{n \rightarrow \infty} x_n = 0 \right\} \quad \text{equipped with norm } \|x\|_{\infty} := \sup_{n \in \mathbb{N}} |x_n|.$$

(i) Prove that $(c_0, \|\cdot\|_{\infty})$ is a Banach space.

(ii) Set

$$Lx := \sum_{n=0}^{\infty} \frac{x_n}{n!} \quad \forall x = \{x_n\} \in c_0.$$

Prove that $L \in c'_0$. Find $\|L\|_{c'_0}$ and prove that this norm is not achieved in c_0 .

The next lecture will be on the equivalence of norms and its consequences.